XE

XELOR · ENGINEERING HANDOFF

MVP Technical Handoff & Implementation Brief

What XELOR is, how it handles the complete operating loop, what was changed, how the investor demo works, how to rebuild and verify it, and what the technical team should do next.

Snapshot: investor-demo build based on the live Northstar PX-400 workflow, additive Factory Connect simulator, RELAY managed-service model and ACHILES private platform assurance.
Audience: engineering, product, QA, service operations, DevOps and technical leadership.
Repository: `MVP_PROTOTYPE_1` · working-directory analysis updated on 8 August 2026.

Prepared from the application, migrations, API contracts, deployment files, supplied deck, primary standards and automated checks

8 August 2026

1. Handoff summary

The MVP was changed from a broad collection of screens and fictional presenter stories into one controlled, evidence-backed investor demonstration. The central story is a real seeded customer commitment that crosses Sales, Planning, Purchase, Inventory, Production, Quality, Maintenance, Employee Spend, Accounts and Agent OS without pretending that unfinished production features are complete.

> **Current result:** the build is demo-ready, deterministic, locally reproducible and protected by automated evidence. The application is not being represented as production-complete; its remaining gaps are named in this document.

99 / 99	32	783 / 783	22
live acceptance checks	browser tests defined	unit and service tests pass	registered web modules

The product claim now demonstrated

XELOR can connect one operating decision across departmental records, calculate deterministic facts, show the evidence, require a named human decision and retain an attributable audit trail.

The product claim deliberately not made

The local build does not claim autonomous external execution, eight independent LLMs, a staffed managed service, production cloud readiness, complete quotation/AP/ECO workflows or AI authority to approve its own work.

Document map

2. What XELOR is and how it handles the whole operating loop

XELOR is a manufacturing intelligence and ERP operating system. It keeps the normal systems of record—customers, orders, materials, production, inspections, assets, people and accounting—but adds a governed coordination layer that recognises when records in different departments are actually part of one business decision.

> **In one sentence:** XELOR turns a factory event into a permission-checked, evidence-linked decision; lets deterministic software calculate the facts; lets AI organise and explain them; requires a human for consequential approval; and records the result for audit and learning.

The problem XELOR is designed to solve

A conventional ERP stores departmental truth but usually makes people assemble the operating picture through calls, spreadsheets and experience. XELOR keeps those modules separate for ownership and control, connects their evidence through ONYX, and uses RELAY to coordinate the human managed service around the technology.

The nine-agent operating model

Agent	Business responsibility	What it contributes	What it cannot do alone
ONYX	Cross-company coordinator	Recognises one decision, scopes the mission, joins evidence and prepares the decision brief.	Cannot grant itself permissions or approve governed work.
MICA	Sales and manufactured-product care	Customer commitment, order, credit context, product case, warranty, spares and commercial options.	Cannot bypass credit, promise unsupported dates or own an operational incident with XELOR itself; that belongs to RELAY.
SPAR	Purchase and inventory	Supply, stock, batch position, supplier options and material movement evidence.	Cannot invent stock or silently change reservations.
AXLE	Engineering and planning	BOM, routing, MRP, capacity, dependency and schedule calculations.	Cannot treat an AI narrative as a planning calculation.
KILN	Production, quality and maintenance	Operation execution, inspection, quarantine, NCR/CAPA, equipment and downtime evidence.	Cannot release failed material or self-certify CAPA effectiveness.
RASP	Finance, working capital and spend	Posted value, budget position, cash/credit context and financial consequence.	Cannot call an unsupported estimate a verified loss.
HEXA	Governance and compliance	Authority checks, policy, approval boundary, audit and verification requirements.	Cannot replace the accountable human approver.
RELAY	Managed Services	Incident clocks, customer updates, service change coordination and improvement evidence.	Cannot diagnose or repair the affected technical component.
ACHILES	Private Platform Assurance	Hourly availability, latency, freshness and append-only check history.	Cannot dispatch work, restart services, change ERP data or contact customers.

How XELOR handles a business event from start to finish

1. **Record:** a human, machine, integration or business transaction creates a normal ERP record such as an order, inspection, downtime event or receipt.
2. **Validate:** request schemas, permissions, idempotency keys and database constraints refuse invalid or duplicated work.
3. **Calculate:** deterministic engines calculate tax, stock, MRP, dates, limits, disposition, balances and policy outcomes.
4. **Detect:** Decision Commander and Agent OS identify that records across departments now form one risk or decision.
5. **Gather:** specialist agents read only the evidence allowed by the user's tenant, role and module permissions.
6. **Explain:** ONYX assembles the facts, causes, options, owners and evidence links. It distinguishes verified value from estimated exposure.

7. **Approve:** if the proposal can create a meaningful consequence, a named person reviews the exact boundary and writes an approval or rejection note.
8. **Execute:** approved internal work is written through the owning domain service, with idempotency and an attributable ancestor.
9. **Propagate:** the transaction writes an outbox event; the worker relays it without making the original ledger write dependent on an external call.
10. **Verify and learn:** outcome records and hash-linked audit evidence show what happened, who decided and whether the expected result was achieved.

ERP signal owned by a module	→	Rules + agents connect evidence	→	Human decision then verified action

The four layers the team must keep separate

Layer	Owns	Examples	Engineering rule
1 · Transactional core	The legal and operational record.	Orders, stock ledger, inspections, production operations, vouchers and employees.	Only the owning domain service writes it; database constraints and RLS remain authoritative.
2 · Decision intelligence	Recognition, evidence joining and options.	Decision Commander, ONYX brief, agent evidence packets and exposure estimates.	May read and propose within permission scope; must cite source records and label uncertainty.
3 · Governed execution	The exact approved work boundary.	Approval inbox, decision note, specialist work item and idempotent mutation.	No consequential action without an authorised ancestor; retries must not duplicate effects.
4 · Proof and learning	What happened and whether it worked.	Audit chain, action ledger, CAPA effectiveness, outcome metrics and reconciliations.	Observed outcomes stay separate from forecasts; verification needs evidence and a named reviewer.

What XELOR does not replace

- It does not replace the domain engines that calculate tax, MRP, inventory, payroll or accounting.
- It does not replace accountable managers, inspectors, engineers, buyers or finance approvers.
- It does not turn an AI response into source data; ERP records and evidence remain the truth.
- It does not imply an external action occurred unless a governed connector confirms and reconciles it.

How the Northstar example demonstrates the model

Northstar's order starts in MICA's Sales records. AXLE calculates material/capacity needs. SPAR supplies and traces material. KILN executes four operations, records a failed inspection, quarantines twelve units and connects Furnace 02 downtime. RASP follows the credit and accounting consequences. HEXA prevents the recovery proposal and CAPA from approving themselves. ONYX presents the entire case as one decision while every department remains owner of its own source record.

> **The key technical distinction:** XELOR is not one chatbot sitting above an ERP. The ERP transactions, permissions, calculations and constraints remain authoritative. The intelligence layer is allowed to read, connect, explain and propose within those boundaries; it does not become a second system of record.

2A. Module atlas — ONYX intelligence and coordination

These three modules are often described together as “the AI layer”, but they have different jobs. Copilot answers one bounded question, Agent OS coordinates a multi-step decision, and AI Operations controls whether model-assisted features are allowed to run.

Module	Purpose and core records	Actual workflow	Controls and current boundary
ONYX Copilot	Read-only factory question surface. Stores 21 known intents and a log of every question, matched intent, outcome, sources, row count and correlation id.	Rules match first; optional model routing may only select a known intent. The user permission is checked, one hand-written query runs inside the tenant transaction, a deterministic answer is rendered, optional narration is checked for numeric provenance, and the event is logged.	No SQL generation or write tool; per-intent permission and row cap; unknown questions refuse; narration falls back safely. The 21 questions are live, but unrestricted conversation and general business advice are not.
Agent OS	Durable cross-functional decision runtime. Persists agent/capability catalogue, graph hash/version, runs, node attempts, events, checkpoints, approvals, signals, outcomes and governed work items.	Freeze tenant/user/graph context; run all ready nodes in parallel waves; checkpoint; join evidence; ask HEXA to verify; pause for the named person; resume; publish the result or append only the approved work item.	Allow-listed tools intersect with the user's permissions; bounded steps, retries and timeout; content-hashed graph; engine checks approval ancestry before an effect. Seven graphs and nineteen capabilities are live; reasoning is deterministic and a work item is not completed ERP or machine work.
AI Operations	Governance console for registered AI features, providers, kill state, tenant opt-outs, budgets, usage/cost events, evaluations, human review, incidents and the AI action chain.	Caller names a feature; router checks registration, global/tenant policy and budget; provider or fallback runs; usage and outcome are logged; golden-set evaluation and human feedback decide eligibility.	Closed registry, kill switch, budget ceiling, privacy mode, deterministic fallback and no-ship evaluation rule. Governance plumbing is real; the demo does not configure a live external provider and some features still lack implementation or golden sets.

How the three fit together

Copilot one known question	→	Agent OS one governed mission	→	AI Operations policy, cost and evaluation

Engineering rule: no one should add a hidden general-purpose prompt or action endpoint to connect these modules. New behavior must enter through a registered intent, capability, graph and permission so the existing refusal, audit and evaluation controls remain meaningful.

2B. Module atlas — HEXA control plane

HEXA's modules are used by every other department. They establish the company context, decide and prove authority, and make external failures visible instead of silently losing them.

Module	Purpose and core records	Actual workflow	Controls and current boundary
Organisation	Company/legal identity, GST registrations, addresses, departments, base currency and financial-year document sequences.	Resolve the company and seller registration for the transaction; let tax/statutory modules use that state; allocate the next document number inside the same transaction that creates the document.	Tenant RLS, typed permissions, unique registration constraints and transaction-safe numbering. It is a live foundation, not a consolidation or corporate-secretarial suite.
Administration	Typed permission keys, roles/grants, SoD rules/findings, access preview, incident and privacy clocks, security posture, audit verification/anchors and licence state.	Build roles from catalogue permissions; grant them; scan conflicting role pairs; enforce server guards; append sensitive events; re-hash audit chains; review open statutory clocks and evidence.	Unknown permission trigger, tenant RLS, server-side guards that filter inactive roles/grants, named SoD rules, append-only audit and break-specific verification. Row/field policy is mainly preview-only; approval escalation still needs work.
Integration	Existing connector reliability records plus additive Factory Connect gateways, asset bindings, state/location/dwell submissions and governed command records.	Store idempotent operational-event submissions; expose robot/AMR context; evaluate only closed allow-listed requests against stored state; retain approval, expiry and audit evidence.	The API records a simulator policy result and does not call `apps/edge` or a controller. OPC UA, MQTT and ROS 2 adapters, plus Cisco Spaces and Splunk context integrations, are catalogue/reserved targets requiring separate implementation, gateway identity and site safety validation.

Verified identity
tenant + actor
→
Permission / approval
exact boundary
→
Audit / integration
evidence and failure state

Important deployment distinction: the normal local mode uses Keycloak OIDC. The public investor deployment can deliberately enable a fixed, sign-in-free demo persona against an isolated seeded database. That mode is an explicit demo exception and must never protect real customer data.

2C. Module atlas — MICA sales and product care; SPAR supply

Module	Purpose and core records	Actual workflow	Controls and current boundary
Sales MICA	Customers, GST/ship-to data, orders/lines, credit snapshots and overrides, delivery notes, invoices and receivables.	Create/check customer; draft order; derive GST from supplier/place-of-supply; confirm through credit gate; dispatch only stock that exists; write delivery, stock-out, invoice and receivable atomically.	Duplicate evidence, immutable credit snapshot, mandatory override reason, server tax, credit/stock refusal and idempotency. Order-to-dispatch is live; lead, quote/tender, reservation and cancellation depth are missing.
Customer Care & Warranty MICA	Customer accounts/users, installed base, warranty/AMC, manufactured-product cases/comments, response-target pauses/breaches, complaints, spares, knowledge articles, drafts and feedback.	Open and triage a product case; compute the business-time response target; check entitlement; assign; prepare a suggestion; require staff review before send; capture complaint, spare and satisfaction outcomes.	Tenant plus customer-account RLS, business-time recomputation, immutable sent comment and refusal of unsupported promises. Product-case and portal controls are live; outbound transport and automatic warranty population are not. XELOR technology incidents stay with RELAY.
Purchase SPAR	Vendors, duplicate candidates, purchase orders/lines, W1 approvals/decisions and goods receipts.	Create/check vendor; draft PO; submit to version-pinned approval route; let only the resolved approver decide; receive only approved outstanding quantity; post receipt and stock together.	Duplicate evidence, named approver, idempotency, approved-only receipt, over-receipt refusal and atomic stock posting. Vendor-to-GRN is live; supplier invoice, three-way match, AP and payment are not.
Inventory SPAR	Warehouses, append-only movements, reference/idempotency key, batch/expiry and row-locked on-hand balances.	Owning module requests a movement; Inventory validates, serialises competing issues, chooses eligible FIFO batches, locks balances, refuses negative stock, appends ledger and updates balance atomically.	One stock write path, advisory/row locks, negative-stock refusal and no-update/no-delete ledger. On-hand is live; bins, reservations, cycle count, serial genealogy and full valuation are not.

Critical cross-module transactions

- **PO receipt:** approved Purchase quantity and Inventory movement commit together; a failed movement cannot leave a completed receipt.
- **Customer dispatch:** Inventory stock-out, Sales delivery/invoice and Accounts receivable commit together; credit and stock gates run before the effect.
- **Service isolation:** customer portal rows carry a restrictive account policy in addition to the tenant fence.

2D. Module atlas — AXLE planning and KILN plant execution

Module	Purpose and core records	Actual workflow	Controls and current boundary
Engineering AXLE	Items, UOM/type/status, tax/planning attributes, BOM headers/versions and component quantities/scrap.	Define item; create/version BOM; validate graph; expose product structure to Planning; let Production pin the structure when work is created.	Tenant/permission controls and cycle detection exist. Item/BOM foundation is live; ECR/ECO intent, impact, approval and effectivity are missing.
Planning AXLE	Policies, calendars, forecast/demand/MPS, immutable MRP workings, planned orders/pegs, exceptions, requisitions, capacity and draft schedule.	Compute levels; consume forecast; net by bucket; protect safety stock; apply lot rules; offset working days; explode into release buckets; persist explanation/pegs; rank exceptions; optionally firm/convert.	Cycle/calendar refusal, immutable completed run, round-up lot sizing, past-due flag and execution uniqueness. MRP is live; capacity is infinite and scheduling remains a draft heuristic.
Production KILN	Orders, BOM/component snapshot, operation sequence/state, operator/evidence, material issue and finished receipt.	Create and pin components; define/start/complete predecessor-gated operations; issue through Inventory; require routed work complete; receive finished quantity; expose execution history.	Idempotency, predecessor gates, accountable evidence and all-blockers completion response. Operations are live; auto-armed quality gate, scrap movement, genealogy and detailed costing are incomplete.
QMS & Audit KILN	Specification/sampling snapshots, inspections/readings/verdicts, dispositions, findings/NCR, CAPA/actions/effectiveness, documents, training, audits and packs.	Choose/store sample derivation; snapshot applied limits; record readings; calculate verdict; quarantine/release; investigate finding; complete actions; let a person judge effectiveness; version evidence pack.	Immutable limits, critical-defect override, movement-linked disposition and human CAPA effectiveness. Core inspection/NCR/CAPA is live; auto-created inspections, calibration and full genealogy are missing.
Maintenance KILN	Assets, readings, requests, work orders/tasks/labour/safety/spares, downtime corrections/disputes, PM occurrences, KPI snapshots and linked Factory Connect evidence.	Acknowledge/triage request; assign and work the job; record asset downtime to handback; enforce close gates; generate PM; calculate reliability from source intervals.	Simulator events add context but do not constitute certified condition monitoring. Shift calendars, predictive models and physical adapters remain future work.

Why this group matters: AXLE states what should happen; KILN proves what did happen. The system must never silently replace execution evidence with a plan, or quality/reliability arithmetic with an AI explanation.

2E. Module atlas — RASP finance, spend and people

Module	Purpose and core records	Actual workflow	Controls and current boundary
Accounts	Chart, balanced vouchers/lines, posting date/status/reference, reversal links, receivables and receipt allocations.	Validate/post voucher; deferred database check re-sums at commit; create receivable from invoice; allocate receipts oldest first; correct with a linked opposite voucher.	Triple balance checks, append-only posting, reversal-not-edit, idempotency and dated reporting. Ledger/trial balance/receivables are live; period close, AP, bank reconciliation and statements are incomplete.
Employee Spend	Budget reservations, travel/claims/lines, attachments/extraction decision, advances, indirect expenses, posting queue, TDS/ITC and duplicate reports.	Lock/check budget; create request; attach/confirm receipt; submit and reserve in-approval; approve/reject; move signed reservation rows to committed/actual; prepare accounting posting.	Atomic reservation, unbudgeted warning, written override, human extraction confirmation and tax gates. Core controls are live; spend-to-ledger posting and cost-centre alignment are incomplete.
People	Employees, grade/bank/statutory data, punches, attendance/regularisation, leave, payroll/payslip, dated rates and posting reference.	Derive days from punches; hold incomplete days; append correction/replay; calculate deemed wage and dated statutory deductions; separate prepare/approve; post payroll to Accounts.	No one-punch shortcut, dated rate lookup, missing-rate refusal, separation of duties and encrypted sensitive fields. Foundations are live; recruitment/performance/learning and device transport are not.
Working Capital	Decision workspace for money in/out, stock cash, margin, 13-week outlook, scenarios and finance-pack gaps; current cash evidence starts from Accounts.	Read posted ledger; add Sales commitments and stock; label assumptions; compare bounded scenarios; prioritise follow-up; draft evidence manifest; pass through HEXA and human approval.	Read/analyse/simulate/draft only, no payment/posting, explicit boundary notes. Cash-position read and mission are live; most workspace values and three finance tools remain illustrative/evidence wrappers.

Accounting truth versus decision aid: a posted voucher and dated trial balance are verified facts. A cash scenario, collection priority or margin explanation is a decision aid until its inputs, calculation and outcome are implemented and reconciled. RASP must keep those labels visible.

2F. Module atlas — RELAY Managed Services

RELAY is not another technical support queue. It is the service-management layer that makes XELOR supportable after connection: one catalogue, one transition, one service incident, one customer change calendar, one review and one improvement register.

Design
outcomes + SLO → **Transition**
runbooks + acceptance → **Operate**
incident + change → **Improve**
review + action

Area	RELAY owns	Specialist/human owns	MVP boundary
Service design & transition	Catalogue, coverage, severity, SLO/SLA schedule, responsibility matrix, discovery, runbooks, acceptance and hypercare.	Specialist validates its domain; human approves commercial/legal terms.	Operating model is implemented; no live customer entitlement or acceptance database yet.
Incidents	Impact, severity, response clock, escalation, timeline, next update and verified service outcome.	Affected specialist diagnoses and repairs; HEXA alone decides security/reportability.	Seeded examples, not live monitoring, paging or ITSM.
Changes & releases	Customer calendar, collision/readiness check, notice and post-change service verification.	Specialist designs/executes; HEXA/human approves where required.	No automated notices or deployment integration.
Review & improvement	Service-level evidence, repeat-failure/problem follow-up, customer review and improvement register.	Specialist implements improvement; human approves scope, credit and contract.	Displayed measures and review are illustrative, not contractual.

Implemented code path

Permission
`overview.read` → **API**
`/managed-services/overview` → **5 web screens**
explicit demo label → **Assurance graph**
HEXA + human gate

Non-duplication rule: MICA keeps product support/warranty; KILN keeps factory maintenance; HEXA keeps security/connectors; ONYX keeps AI operations and business missions. RELAY keeps the service clock and customer hand-off around them.

> **Production work still required:** tenant-scoped service tables and RLS; OpenTelemetry ingestion; ITSM/paging/customer-channel adapters through HEXA Integration; staffed coverage/on-call; measured SLOs; restore and escalation drills.

3. Design principles and before/after

The hardening work followed four rules. These rules should remain release criteria for future changes.

One business story plus one agent overview

The primary demo follows a simple customer-order-to-delivery journey through real ERP screens. A separate overview opens all nine agent maps and explains only each role and hand-off, so it does not interrupt the business story.

Persisted facts, not narration

Named quantities, dates, values, operators, readings, tasks and approvals exist in the database and are read through authenticated APIs. Illustrative screens identify themselves.

AI coordinates; people decide

ONYX and specialist agents gather and connect evidence. A graph cannot reach governed action nodes until an authorised human approves the exact boundary with a recorded note.

Failure must be understandable

Invalid input, duplicate requests, missing idempotency and insufficient stock return controlled responses. The demo does not depend on hiding server errors.

Area	Before hardening	Current implementation
Presenter journey	Thirteen selectable scenarios, many based on narration-only records.	Exactly two guided choices: an 11-step real factory journey and a 9-step high-level agent overview. Both use existing product screens.
Production	Production order stored only an overall quantity/status.	Four ordered operations with predecessor rules, operator, timestamps, output/reject quantity and evidence.
Quality	Complaint fields could contain NCR/CAPA text references, but no quality workflow records existed.	Persisted NCR, containment, root cause, CAPA completion and separately controlled effectiveness verification.
Batch stock	Batch receipts could be visible but unnamed issues could not reliably consume them.	FIFO allocation locks eligible batches, records explicit splits and refuses insufficient aggregate stock before posting.
Employee Spend	Extensive backend existed but no registered web module exposed it.	Live claims and budget views appear as the nineteenth module.
Credit decision	Decision-time snapshots were written but not visible to API/UI users.	Limit, prior exposure and exposure including this order are displayed separately.
Demo provenance	Some curated UI could be mistaken for live calculations.	Live, illustrative, deterministic and human-owned boundaries are labelled in the interface.
Inactivity	Thirty seconds without input navigated an authenticated user back toward the gateway/sign-in.	The idle redirect and its unused form-hold hooks are removed. Sessions refresh normally and screens remain in place.

4. Runtime architecture

Next.js 15 / React 19 **authenticated web app** `:3001`	→	**NestJS 11** **modular API** `:3000/api/v1`	→	**PostgreSQL 17** **system of record** `:5432`
Keycloak 26 **OIDC + PKCE** `:8080`	→	**Tenant + RBAC guards** **147 registered permissions**	→	**FORCE RLS** **233 tenant tables**
Transactional outbox **domain events**	→	**Valkey + worker** `:6379`	→	**Idempotent consumers** **attributable effects**
Deterministic rules **and evidence reads**	→	**ONYX / Agent OS** **decision coordination**	→	**Human approval** **before governed action**

Component responsibilities

Component	Responsibility	Important implementation boundary
`apps/web`	Product shell, 22 modules, presenter guide, ONYX, RELAY and ACHILES surfaces and authentication callback.	UI permission checks improve usability; API guards remain authoritative.
`apps/api`	Domain services, controllers, Agent OS, audit, validation and worker entrypoint.	Ledger and workflow writes remain inside tenant-scoped transactions.
`apps/edge`	Standalone factory adapter contract, process-local duplicate suppression, expiry checks, closed capabilities and an explicit simulator.	It is not wired to the Factory Connect API and has no durable edge journal or claim/acknowledgement transport. No physical controller connection ships in the MVP; local safety control remains final authority.
`packages/platform`	Permissions, error types, deterministic business rules, Agent OS contracts.	No browser or database coupling; pure rules remain unit-testable.
`packages/db`	Drizzle schema, SQL migrations, tenancy checks, reset and migration tooling.	PostgreSQL constraints and FORCE RLS defend invariants below the API.
Keycloak	Real OIDC tokens, tenant/persona claims and local presentation theme.	Arbitrary credentials are rejected; the known presenter shortcut is demo-only.
Worker	Relays outbox events and executes idempotent demo consumers.	API success does not depend on silently dropping asynchronous work.
Gotenberg	Local document-rendering service available on `:3002`.	It is infrastructure availability, not a claim that every module exports governed PDFs.

> **Architecture choice:** keep the modular monolith through product-market fit. Manufacturing transactions need strong cross-domain consistency; separate web, API and worker processes provide operational scaling without introducing premature distributed transactions.

5. Canonical Northstar dataset

The demo is anchored to one commitment rather than a collection of disconnected examples. The following facts are seeded, read back through APIs and asserted by the acceptance matrix.

Customer promise

Northstar Process Systems
Customer PO: `NPS/PO/10482`
120 × PX-400 pumps
₹74.34 lakh
Promise: 04-Sep-2026

Current position

40 produced
28 passed and dispatched
12 quarantined
80 still remaining
Finished-goods balance: 0 after dispatch

Control position

Credit override has snapshots
Inspection failed on evidence
CAPA not self-closed
Human approval required
Audit chain remains verifiable

Business record	Seeded fact	Why it matters
Sales order	120 PX-400, ₹74.34 lakh, IGST, 04-Sep-2026; 28 dispatched and 80 remaining.	Defines the customer commitment and commercial exposure.
Credit decision	Limit snapshot and existing exposure preserved separately from order-inclusive exposure.	Shows what the human knew at the decision point.
Planning run	Three-level BOM explosion with material and capacity exceptions.	Dates and shortages are deterministic planning outputs, not AI guesses.
Production order	Four completed operations; 40 units received.	Proves accountable shop-floor execution.
Inspection	0.034 mm shaft runout against a 0.020 mm limit; rejected.	Connects measured evidence to product disposition.
Disposition	28 passed units available for dispatch; 12 moved to quarantine.	Prevents “produced” from automatically meaning “usable”.
NCR/CAPA	`NC-2627-00001` and `CAPA-2627-00001`; action complete, effectiveness pending.	Shows separation between doing corrective work and proving that it worked.
Maintenance	Furnace 02 vacuum failure, 4.5 hours downtime, three signed tasks, ₹1,855.14 labour.	Connects the constrained asset to delivery feasibility.
Employee Spend	Witness-test travel claim with three lines and an FY 2026-27 budget decision.	Makes employee spend visible inside the product rather than API-only.
Accounts	Partial dispatch created a posted accounting voucher and trial balance remains balanced.	Links operating execution to recorded finance evidence.

Reconciliation: 40 produced = 28 dispatched + 12 quarantined. No passed PX-400 remains in finished-goods stock after the dispatch. This equality is explicitly tested.

6. The two investor demos

Demo 1 — From customer order to delivery is an 11-step, non-technical factory story. It pauses at the two moments where the presenter should create and show a real document. **Demo 2 — Meet the agents** is a separate 9-step overview that opens every agent map and says only what that agent does and who it hands work to.

Demo 1 — real factory journey

#	Route	What the presenter proves	Primary evidence
1	`/sales/orders`	Enter the customer's request as a real sales order.	The presenter clicks New order, saves it, shows the saved order, and only then presses Next.
2	`/planning/mrp`	Work out what the factory needs to satisfy demand.	Material requirements and proposed dates are calculated rather than guessed.
3	`/purchase/orders`	Order material that is missing.	The presenter creates a real purchase order, shows it, and only then presses Next.
4	`/inventory/stock`	See what is physically available.	Stock remains different from an order or expectation.
5	`/production/orders`	Turn material into finished pumps.	Production work, quantity and status have an accountable owner.
6	`/quality/inspections`	Check that the result is acceptable.	A failed measurement stays visible and the affected quantity remains held.
7	`/agentos/commander`	Connect the delivery risk across departments.	ONYX combines owned evidence into one understandable decision.
8	`/agentos/approvals`	Keep the consequential choice with a person.	The exact action, evidence and decision note are visible before approval.
9	`/accounts/vouchers`	Connect delivery to recorded money.	Posted finance evidence remains separate from unsupported estimates.
10	`/csp/tickets`	Show MICA's product after-sales responsibility.	Manufactured-product cases, warranty, spares and feedback stay with MICA.
11	`/managed-services/responsibilities`	Show RELAY's separate technology-service responsibility.	XELOR application, integration and AI incidents are coordinated by RELAY, with technical repair retained by the right specialist.

Demo 2 — meet the nine agents

The guide visits `/department/ONYX`, `HEXA`, `MICA`, `SPAR`, `AXLE`, `KILN`, `RASP`, `RELAY` and finally `ACHILES`. Each step opens the agent map normally and provides one plain-language role explanation plus one hand-off line. It intentionally omits technical internals, live-record claims and lengthy evidence notes.

Presenter guide behaviour

- The guide never writes a business record itself. At Sales and Purchase, the presenter deliberately uses the normal real form.
- Saving the required sales or purchase order unlocks Next but never auto-advances, so the presenter can show the saved detail page first.
- While either order form is open, the floating guide temporarily hides so it cannot cover the form footer or intercept the Save action; it returns after the dialog closes.
- A plain “On this screen” line provides context without covering or highlighting the ERP interface; expanded notes provide a simple explanation, facts and a presenter line.
- The business-story dock states that it uses real ERP screens and never saves for the presenter.
- The top control says **Stop demo**, not Reset demo, because it only removes the presentation layer.
- Refreshing or stopping the guide returns the user to normal product use without altering the ERP record.

> **Important distinction:** the guided layer contains explanatory vocabulary. The business quantities and named records underneath are live seeded data. Static future-workspace screens retain a visible “Illustrative workspace” boundary.

7. Domain changes — operational workflows

7.1 Sales and credit evidence

- Sales order views now expose the decision-time `creditLimitSnapshot` and `creditExposureSnapshot`.
- The order screen separately presents credit limit, exposure before this order and exposure including this order.
- Partially dispatched orders remain visible in Decision Commander even when the original commitment is older than the earlier 14-day window.
- The canonical Northstar order remains `part_dispatched`; the demo does not pretend the remaining 80 are complete.

7.2 FIFO batch allocation

When a withdrawal does not nominate a batch, Inventory now locks eligible batch balances, sorts oldest first and creates an explicit allocation split. It validates the aggregate quantity before any ledger posting.

Batch A oldest eligible	→	Requested quantity allocated in order	→	Stock ledger one line per split

- Zero balances are ignored.
- One issue can consume multiple batches and retain each split.
- Insufficient total stock fails before a partial posting.
- Advisory locking prevents two withdrawals from allocating the same balance concurrently.

7.3 Executable production route

Migration `0065_production_operations.sql` adds `production_operation`. The API supports adding, starting and completing operations. Completion checks ordered predecessors, requires an operator and stores quantities plus an evidence note.

Kit verification	→	Rotating assembly	→	Pump assembly then hydro pretest

- Valid states: queued, in progress, completed and blocked.
- Planned and actual time windows are constraint-checked.
- A production order cannot complete while any routed operation is incomplete.
- The production detail UI shows route evidence, operator, time and accepted/rejected quantity.

7.4 Maintenance integrity

- Work-order start and task addition require idempotency keys.
- Employee references, task values and budget queries are zod-validated before service execution.
- The Northstar work order stores three measurable signed tasks, including a 0–0.05 mbar vacuum bound.
- Labour is opened once on start and closed/valued on completion; retries no longer duplicate it.

7. Domain changes — quality, spend and decision intelligence

7.5 Persisted NCR and CAPA

Migration `0064_quality_corrective_workflow.sql` introduces `qms_finding` and `qms_corrective_action`, both tenant-fenced, non-deletable to the application role and protected by database checks.

Rejected inspection → **NCR containment + cause** → **CAPA completion ≠ proof**

- Finding sources include inspection, audit, complaint, supplier and manual.
- Containment must exist after the finding leaves `new`.
- Root cause requires an attributable confirmer and timestamp.
- CAPA completion evidence is required before effectiveness review.
- A CAPA may close only after an effective verdict with evidence, verifier and timestamp.
- The Northstar CAPA intentionally remains in effectiveness review for a human.

7.6 Employee Spend module

- A registered `expenditure` web module now exposes claims and budget decisions.
- The Northstar witness-test claim is visible with three lines and reconciled totals.
- Budget consumption accepts the canonical compact Indian FY code `2627`.
- Malformed budget queries now return a validation response rather than a server failure.

7.7 Decision Commander

- Joins current sales, planning, purchase, quality and maintenance reads.
- Shows source references, dates and values as decision facts.
- Uses explicit trust chips: Live ERP records, Deterministic risk rules, No hidden writes and Human approval required.
- Calls unsupported monetary consequence **exposure**, never verified loss.
- Leaves amounts blank when a defensible rate is unavailable.

7.8 Agent OS and ONYX

- Nine governed identities are registered: ONYX, MICA, SPAR, AXLE, KILN, RASP, HEXA, RELAY and read-only ACHILES.
- Evidence waves may run in parallel, but action nodes sit behind a durable human checkpoint.
- Approved actions are internal accountable work items; no external supplier, payment or customer system is claimed to have been contacted.
- One PostgreSQL client-concurrency warning was removed by making tenant-pinned Agent OS reads sequential inside their transaction.

8. Database and API changes

New persisted structures

Migration / schema	Main structure	Key invariants
`0064_quality_corrective_workflow.sql`	`qms_finding`, `qms_corrective_action`, NC/CAPA document series.	Tenant uniqueness, idempotency, containment/cause/closure checks, separate effectiveness verification, FORCE RLS, no application DELETE.
`0065_production_operations.sql`	`production_operation` linked to `production_order`.	Unique sequence per order, valid time windows, accountable started/completed states, FORCE RLS, no application DELETE.
Production Drizzle schema	Typed operation relation returned with order detail.	API and UI compile against one field definition.
Quality Drizzle schema	Typed finding and CAPA records.	Status and effectiveness values remain constrained in TypeScript and SQL.

New/extended API surface

Method and route	Purpose	Control
`GET /quality/findings`	List persisted NCR findings.	`quality.inspection.read`
`POST /quality/findings`	Create a source-linked finding.	Execute permission + idempotency + zod
`POST /quality/findings/:no/contain`	Record containment.	Disposition permission + idempotency
`POST /quality/findings/:no/root-cause`	Confirm attributable root cause.	Disposition permission + idempotency
`GET/POST /quality/corrective-actions`	List or create CAPA.	Read/decision permission as appropriate
`POST /quality/corrective-actions/:no/complete`	Store completion evidence.	Does not close effectiveness
`POST /quality/corrective-actions/:no/verify`	Record effective/ineffective human result.	Evidence + attributable actor
`POST /production/orders/:id/operations`	Add an ordered operation.	Create permission + idempotency
`POST .../operations/:sequence/start`	Start with operator and optional input quantity.	Execute permission + predecessor checks
`POST .../operations/:sequence/complete`	Complete with output, rejects and evidence.	Execute permission + validation
`GET /production/orders/:id`	Return order and operation route.	Read permission + tenant scope
`GET /integration/factory/overview`	Return gateway, robot/AMR, location, dwell and command evidence.	`factory.connect.read` + tenant RLS
`POST /integration/factory/telemetry/state`	Append an idempotent asset-state submission.	`factory.telemetry.ingest` + source-event idempotency key; no cryptographic plant-gateway identity is claimed

Method and route	Purpose	Control
`POST /integration/factory/commands`	Evaluate and record one closed, expiring simulator intent linked to a completed approved Factory Flow gate.	`factory.command.execute`; exact intent match and one-time approval consumption; no edge dispatch or controller acknowledgement

Cross-cutting response behaviour

- Mutations in the changed paths require `Idempotency-Key`; missing keys fail before writes.
- Direct and ORM-wrapped PostgreSQL `23505` conflicts map to a safe canonical HTTP 409 response.
- Zod failures produce field-specific validation output instead of positional-field 500 errors.
- Unknown production orders return clean not-found responses.

9. Identity, security and governance

Authentication

- The application uses real Keycloak OIDC Authorization Code + PKCE.
- The local presenter identity is `hari`; arbitrary credentials no longer become an administrator.
- Blank fields are allowed only as an explicitly labelled local presenter shortcut and resolve to the seeded demo account.
- Malformed bearer tokens and unauthenticated reads are rejected.
- The browser refreshes the access token before expiry.
- The former 30-second inactivity redirect is removed. Inactivity no longer navigates away or discards presentation context.

Authorisation and tenancy

Verified Keycloak token tenant + subject	→	Route permission guard deny by default	→	PostgreSQL SET LOCAL FORCE RLS policy

- 147 permissions are registered and reconciled against the routes that demand them.
- The permission catalogue covers two seeded tenants with no uncatalogued grants.
- 233 tenant-scoped tables pass the FORCE RLS/policy check.
- The stores persona is explicitly denied the administration role registry.
- UI navigation contains 163 permission references and fails validation if a module invents an undefined permission.

Human and audit controls

- Approvals retain actor, time, written note, exact proposed consequence and outcome.
- Approval grants only the exact action boundary; it is not a broad agent privilege.
- Audit and AI action chains are append-only and hash-linked.
- CAPA effectiveness, credit override and Agent OS action gates remain human-owned.
- “No hidden writes” is a product rule: the guide never saves for the user, while the two clearly labelled order-entry gates use normal ERP forms and remain presenter-controlled.

> **Production warning:** local demo credentials, local realm configuration and development infrastructure must never be reused as production security configuration. Production needs MFA, secret management, WAF, restore drills, monitoring and a formal threat model.

10. Web product and presentation changes

Module architecture

- Employee Spend is registered as module 19.
- Quality now exposes live findings and corrective actions beside inspections.
- Production order detail renders the operation route and evidence.
- Sales order detail renders the three decision-time credit facts.
- Module isolation tooling prevents cross-module imports and validates manifests.

Investor trust design

UI change	Reason
Decision Commander title changed to “Live operating decision room”.	Frames the screen around current operating evidence rather than an abstract dashboard.
Trust chips label live records, deterministic rules, no hidden writes and human approval.	Makes the operating boundary visible without requiring technical explanation.
Static future workspaces show an “Illustrative workspace” banner.	Prevents a curated scenario from being mistaken for a ledger calculation.
One Northstar story replaces the old catalogue.	Removes contradictory fictional numbers and makes the pitch memorable.
Guided dock displays live record reference and step-specific evidence.	Lets the presenter point to proof instead of reading generic narration.
“Stop demo” replaces “Reset demo”.	The button stops navigation guidance; it does not clear business data.
Copilot Alerts/Brief/Actions link to current governed screens.	Removes permanent sample cards from the normal product shell.
30-second screen saver removed.	An investor or operator can pause on a screen without being returned to sign-in.

Login behaviour

The cinematic sign-in remains responsive on desktop and phone, supports dark mode and reduced motion, and uses real Keycloak access. Returning deliberately to the root/sign-out path still requires credentials; ordinary inactivity no longer does.

> **Presentation safety:** the normal shell stays operational underneath the guide. Stopping, finishing or refreshing removes the guide layer and does not alter the business record.

11. Deterministic seed and rebuild process

The demo database can be rebuilt from source rather than repaired manually before each meeting.

Guarded demo reset **drop/recreate schema**	→	**87 migration files** **through 0088**	→	**Base world** **54 successful steps**
Northstar story **99 verified checks**	→	**Worker drains** **transactional outbox**	→	**99-check matrix** **must pass**

Normal rebuild commands

```
corepack enable
pnpm install
pnpm db:demo-reset --yes
pnpm db:migrate
pnpm demo:seed
pnpm demo:northstar
pnpm demo:verify
```

Seeder guarantees

- The reset tool validates the target database and requires an explicit demo reset flag.
- Base and Northstar seeders check every expected response rather than silently continuing after failure.
- Maintenance task mutations carry idempotency keys and no longer multiply labour/task rows on retries.
- The quality action completes but is intentionally left before human effectiveness closure.
- Audit and AI chains are verified at the end of seeding.
- The final story facts are asserted again by API after browser testing.

> **Do not reset a shared or production database.** The demo reset intentionally drops and recreates the public schema. Keep the target isolated, inspect `DATABASE_URL`, and rely on the guard rather than bypassing it.

12. Verification and acceptance evidence

Live investor acceptance matrix — 99/99 on 6 August 2026

Check family	Examples	Result
Runtime and identity	Web, Keycloak realm, OIDC discovery, admin/stores tokens, unauthenticated and malformed-token rejection.	PASS
Authenticated surfaces	32 operational reads across masters, orders, stock, QMS, spend, accounting, Copilot, Agent OS, knowledge graph and administration.	PASS
Northstar commercial facts	120 units, ₹74.34 lakh, 04-Sep-2026, 28 dispatched, 80 remaining and credit snapshots.	PASS
Production and quality	Four complete operations, 40 received, 0.034/0.020 rejection, 12 quarantine, 28 + 12 reconciliation.	PASS
NCR/CAPA and maintenance	Inspection link, non-self-closing CAPA, Furnace 02, accountable work order and 4.5-hour downtime.	PASS
Spend and finance	Expense claim reconciliation, employee attribution, posted voucher and balanced trial balance.	PASS
Decision and governance	Five source families, partially dispatched promise, exposure labelling and nine registered agents.	PASS
Negative/error paths	Persona denial, invalid pagination, missing idempotency, unknown order and malformed spend query.	PASS

Browser verification — 32 current scenarios

- Administrator traverses more than 40 application routes and uses ONYX evidence.
- All 20 guided steps reach a visible target: 11 business-story steps and 9 agent-overview steps.
- The sales-order and purchase-order gates remain locked until the matching form reports a successful save; saving unlocks Next but never changes the route automatically.
- Agent OS runs the nine-agent controlled-autonomy flow; ACHILES contributes read-only health evidence and the same human approval boundary remains intact.
- Specialist personas see only their departments and the API enforces the same wall.
- Desktop, phone, dark mode and reduced-motion identity variants are covered in the normal Keycloak configuration.
- Unknown credentials are rejected in normal mode; public-demo mode is separately checked for sign-in-free entry and persona scoping.
- A dedicated regression leaves Sales Orders untouched for 32 seconds and confirms no redirect.

Static and policy gates

712	4	56	11
platform tests	database tests	API tests	edge + web tests

- Lint passes with zero warnings.
- TypeScript checks pass in web, API, platform, database and edge packages.
- Production Next.js and NestJS builds succeed.
- Database naming check passes across 243 tables.
- All 22 module manifests and 163 navigation permission references pass validation.

13. Local engineering runbook

Prerequisites

- Node.js 22
- Corepack / pnpm 9
- Docker with Compose support
- Ports 3000, 3001, 3002, 5432, 6379 and 8080 available

First-time setup

```
git clone https://github.com/Harirajiv-web/XELOR-MVP.git
cd XELOR-MVP
corepack enable
corepack prepare pnpm@9.12.0 --activate
pnpm install --frozen-lockfile
cp .env.example .env
docker compose version
pnpm infra:up
pnpm db:migrate
pnpm build
```

Start the application

```
# Terminal 1 — API
pnpm --filter @ind-core/api start

# Terminal 2 — asynchronous outbox worker
pnpm --filter @ind-core/api worker

# Terminal 3 — production web build
pnpm --filter @ind-core/web start

# Terminal 4 — after API and Keycloak are ready
pnpm demo:seed
pnpm demo:northstar
pnpm demo:verify
```

Service	URL / port	Expected check
Web	`http://localhost:3001`	HTTP 200 and XELOR sign-in.
API	`http://localhost:3000/api/v1`	Unauthenticated business read returns 401.
Keycloak	`http://localhost:8080`	Realm discovery returns 200.
Gotenberg	`http://localhost:3002/health`	Chromium and LibreOffice report up.
PostgreSQL	`localhost:5432`	Container healthy; migrations current.
Valkey	`localhost:6379`	Container healthy; worker connected.

Verify before any presentation

```
pnpm demo:verify
pnpm lint
pnpm typecheck
pnpm test
pnpm db:rls-check
pnpm db:perm-check
pnpm --filter @ind-core/db naming-check
pnpm --filter @ind-core/web module-check
pnpm --filter @ind-core/web exec playwright test
```

> **Presenter access:** normal local mode uses Keycloak and the seeded presenter account. The five-container public-demo mode below sets `API_PUBLIC_DEMO=true` and `NEXT_PUBLIC_PUBLIC_DEMO=true`, so the product opens directly with a fixed demo persona and no sign-in page. Its browser identity includes the two document-create permissions required by the walkthrough—`sales.order.create` and `purchase.po.create`—while the API independently rechecks those grants against the isolated demo administrator.

13A. Public demo deployment architecture

The repository now contains a complete Railway-ready five-service package designed for a temporary, sign-in-free investor URL. It is deliberately smaller than local Compose: Keycloak is replaced only in explicit public-demo mode and unused Gotenberg is omitted.

Public browser HTTPS	→	xelor-web Next.js	→	xelor-api NestJS + private DNS

Durable services

xelor-postgres holds the application records and a mounted data volume. **xelor-worker** consumes the transactional outbox. **xelor-valkey** provides Redis-compatible coordination/cache and is not the business source of truth.

Only one public surface

Generate a public Railway domain for **xelor-web** only. Web proxies the API path over Railway private DNS. Database, Valkey, API and worker stay private.

Five GitHub services and supplied config

Railway service	Config-as-code file	Start behavior
`xelor-postgres`	`infra/railway/postgres.railway.json`	PostgreSQL 17 + pgvector; bootstraps the non-owner `app_user`; requires a persistent volume.
`xelor-valkey`	`infra/railway/valkey.railway.json`	Pinned Valkey 8 with health check; disposable in demo mode.
`xelor-api`	`infra/railway/api.railway.json`	Wait for DB → migrate → start API → seed/checkpoint once → expose readiness.
`xelor-worker`	`infra/railway/worker.railway.json`	Wait for API/Valkey → start independent outbox worker.
`xelor-web`	`infra/railway/web.railway.json`	Wait for API health → run Next.js on Railway's dynamic port.

Required configuration boundaries

- Shared secrets: separate strong `POSTGRES_PASSWORD` and `APP_DATABASE_PASSWORD`.
- API: owner/app database URLs, `VALKEY_URL`, `API_PUBLIC_DEMO=true`, `AI_PROVIDER=stub`, `SEED_DEMO_ON_BOOT=true` and a versioned `DEMO_SEED_VERSION`.
- Web: `NEXT_PUBLIC_PUBLIC_DEMO=true`, the private API origin and API health URL. This setting is baked into the web build as well as present at runtime.
- The public-demo API still requires its dedicated header and maps only known seeded personas; unauthenticated ordinary API calls continue to return 401.

GitHub-to-public-URL flow

Push selected branch	→	Railway watch pattern builds changed service	→	Health check passes deployment replaces prior

> **Cost and production warning:** five services fit Railway's trial topology but are unlikely to remain inside the long-term free allowance. Suspend the project between demos. This setup has disposable demo data, no HA/restore/SLO claim and must not carry real company data.

> **Custom subdomain:** after the Railway URL works, add `xelor.kisancred.com` to `xelor-web`, copy Railway's exact CNAME target into the authoritative DNS provider, wait for TLS, and keep the generated Railway domain as fallback.

14. Eight-to-ten-minute presentation script

1. **Open the Brain.** Say: “9 governed agents are connected; this local build uses deterministic reasoning so we can prove every result.”
2. **Press Start Demo → From customer order to delivery.** Say: “We will follow one ordinary customer request through the whole factory.”
3. **Sales.** Click New order, create a simple customer order, save it and show the saved detail. Say: “Nothing moves until I press Next.”
4. **Planning.** Explain that the system works out what must be made and bought instead of asking the audience to understand MRP terminology.
5. **Purchase.** Create the missing-material purchase order, save it, show it, and then press Next.
6. **Stock, production and quality.** Explain available material, the making work and the check in everyday language.
7. **Decision Commander.** Show how ONYX connects the same delivery risk across departments.
8. **Approval.** Show that a person reviews exact evidence and writes a decision note before governed work is dispatched.
9. **Finance.** Connect delivery to the posted accounting evidence without claiming an unsupported loss.
10. **MICA and RELAY.** End with the clean boundary: MICA handles the manufactured product after sale; RELAY coordinates problems with XELOR technology itself.
11. **Optional second demo.** Restart and choose Meet the agents when the audience wants the eight high-level roles and their hand-offs.

Questions the technical team should be ready to answer

Question	Short truthful answer
“Is this data real?”	It is deterministic seeded ERP data persisted in PostgreSQL and read through authenticated APIs; it is not a production customer's data.
“Did AI change the ledger?”	No. ONYX gathers and explains evidence. Governed actions require a human gate, and the demo does not claim external execution.
“Are 9 LLMs running?”	No. 9 governed agent identities and workflows are live; language reasoning is deterministic in this environment.
“How is tenant isolation enforced?”	Verified identity establishes tenant context, route guards enforce permissions and PostgreSQL FORCE RLS provides the final fail-closed boundary.
“Can the demo be rebuilt?”	Yes. Guarded reset, migrations, two seeders and the 99-check matrix reproduce and verify it.
“Is it production-ready?”	No. The core architecture is credible, but cloud operations, security assurance and named workflow gaps remain roadmap work.

15. Known boundaries and unresolved product scope

Boundary	Current truth	Required production work
Opportunity and quotation	Sales starts at a committed order.	Lead/opportunity pipeline, quote revisions, validity, approval and quote-to-order conversion.
Supplier AP/payment	Direct-material purchase cycle reaches goods receipt; employee/indirect spend is separate.	Supplier invoice, three-way match, AP ageing, exceptions and controlled payment runs.
Engineering change control	BOM revision exists, but formal change intent/effectivity is not governed.	ECR/ECO, where-used analysis, approvals, effectivity and old-stock disposition.
Production depth	Operation execution is real for the demo path.	Deeper WIP, labour/machine costing, terminal UX, rework and dispatch sequencing.
QMS breadth	NCR/CAPA is real for the Northstar thread; several document/audit workspaces remain illustrative.	Full 8D, attachments, document lifecycle, audit programme persistence and customer-specific routes.
AI provider	Deterministic stub; no hosted model active.	Provider security/privacy review, eval gates, shadow mode, cost controls and rollback before activation.
Infrastructure	Local Compose, Railway-ready service descriptors, and GitHub verification for clean images plus the public-demo critical path. No live cloud URL is confirmed.	Operate a hosted environment, then extend CI with signed promotion/rollback, managed recovery, secrets, observability, restore evidence, HA and incident response.

> **Do not market around these boundaries.** The demo is strong because it is precise about what it proves. Adding unsupported claims would reduce technical credibility more than adding another screen would increase it.

16. Recommended next engineering work

Next 30 days — turn the demo into a governed release

- Review and commit the current large working tree in small, domain-labelled changes; preserve the two new migrations and acceptance tooling.
- Keep the 99-check matrix, full route crawl and simulator-only Factory Flow journey mandatory in CI.
- Promote the supplied Web/API/Worker/PostgreSQL/Valkey images through CI and retain the API startup migration/readiness gate.
- Introduce structured logs, correlation IDs across browser/API/worker and basic error tracking.
- Move repeated module modals into a shared spine component and retain focus/locked-write behaviour.
- Formalise demo reset/runbook ownership and a pre-meeting checklist.

Days 31–60 — pilot foundation

- Deploy a non-production cloud environment with managed PostgreSQL, managed Valkey, object storage, KMS/secrets and TLS.
- Add backup restore drills, database migration rollback rehearsals and environment promotion.
- Enable MFA, production Keycloak policy, secret rotation, rate limits, WAF and audit export.
- Add observability dashboards for latency, error rate, queue lag, database saturation and approval age.
- Complete accessibility and performance budgets for the core investor/pilot paths.

Days 61–90 — close workflow gaps for a design partner

- Prioritise quotation/opportunity because it is the most visible commercial workflow gap.
- Implement supplier invoice, three-way match and AP ageing before claiming end-to-end purchasing.
- Add formal engineering change control for revision/effectivity traceability.
- Deepen operation costing/WIP and QMS attachment/8D support based on the first design partner.
- Evaluate a hosted AI provider only after deterministic baselines, privacy controls and shadow-mode metrics are approved.

> **Recommended ownership:** one engineer owns demo determinism and acceptance; domain owners own workflow depth; platform/security owners own tenancy, identity, audit and deployment gates. Do not make the demo seeder the only place where cross-domain behaviour is understood.

17. File map and handoff checklist

Primary implementation locations

Area	Primary files/directories
Northstar data	`apps/api/scripts/demo/01-seed-base-world.mjs` `apps/api/scripts/demo/02-seed-northstar-story.mjs`
Acceptance matrix	`apps/api/scripts/diagnostics/investor-demo-matrix.mjs`
NCR/CAPA API	`apps/api/src/modules/quality/qms-workflow.controller.ts` `qms-workflow.service.ts`
Production operations API	`apps/api/src/modules/production/production.controller.ts` `production.service.ts`
FIFO inventory	`apps/api/src/modules/inventory/inventory.service.ts` `inventory.service.test.ts`

Database	`packages/db/migrations/0064_quality_corrective_workflow.sql`
	`0065_production_operations.sql`
	`packages/db/src/schema/quality.ts`
	`production.ts`

Guided demo	`apps/web/src/spine/demo/demo-scenarios.ts`
	`demo-launcher.tsx`
	`demo-presenter.ts`

Decision Commander / Agent OS	`apps/api/src/agent-os/` `apps/web/src/modules/agentos/`

RELAY Managed Services	`packages/platform/src/managed-services/`
	`apps/api/src/modules/managed-services/`
	`apps/web/src/modules/managed-services/`
	`docs/01-agent-os/04-managed-services.md`

Employee Spend UI	`apps/web/src/modules/expenditure/`
Browser verification	`apps/web/e2e/` `apps/web/playwright.config.ts`

Identity/theme	`apps/web/src/spine/auth/`
	`infra/keycloak/realm-indcore.json`
	`infra/keycloak-themes/`

Technical handoff checklist

- ☐ Every engineer can rebuild the demo database and reach 99/99.
- ☐ QA can run the 32-scenario Playwright suite and identify both demos, all 20 steps and both manual order gates.
- ☐ Backend owners understand the new operation and NCR/CAPA state machines.
- ☐ Database owners review the RLS, no-delete and constraint choices in migrations 0064/0065.
- ☐ Frontend owners understand live versus illustrative provenance labels.
- ☐ Security owners replace local presenter credentials before any shared deployment.
- ☐ DevOps owners run API and worker separately and monitor outbox lag.
- ☐ Product owners accept the named quotation, supplier AP and ECO roadmap gaps.
- ☐ Service owners accept RELAY's responsibility boundary and the explicit illustrative-data label before any customer demo.
- ☐ The team keeps the inactivity-redirect regression test so the 30-second behaviour does not return.
- ☐ No one claims external autonomous execution or production-grade hosted AI from this build.

> **Definition of a safe handoff:** another engineer can start the stack, rebuild the data, run every gate, present the Northstar story and explain each product boundary without relying on undocumented knowledge from the original implementer.

Companion documents: `docs/02-investor-demo/01-presenter-walkthrough.md`, `docs/02-investor-demo/02-capability-gaps.md`, the technology roadmap report and the Agentic AI implementation strategy.